… Lecture 8 …
Genomic Evaluation

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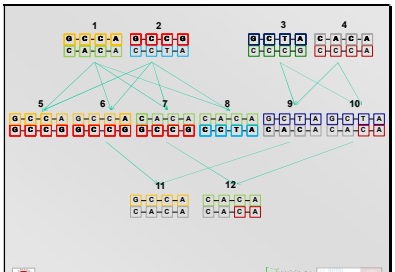
Part 1: Looking at **G** matrices

First we are going to look at the calculation and characteristics of a **G** matrix in a toy example.



The pedigree shown is the same as the one in Lecture 6 with markers at 0, 10, 20 and 30 cM. The unknown QTL at 12 cM has been omitted, and since its genotypes were unknown it would not contribute to calculating **G**. Perhaps the first things to note is that the genomic relationship matrix **G** is independent of **(i)** the exact positions of the marker loci and **(ii)** the phases of the marker alleles of the *m* genotypes for each individual are irrelevant. What is being assumed is that the markers are dense enough to capture all the genetic variance. Although *m* = 4 markers is very small for this assumption it does make it feasible to show something!

Allele Summary

	Locus 1	Locus 2	Locus 3	Locus 4
Reference allele	G	A	T	G
Alternative allele	C	C	C	A
Frequency of alternative, p_i	13/24	15/24	19/24	19/24
Mean {0,1,2} coding, $2p_i$	1.08	1.25	1.58	1.58
Variance {0,1,2} coding, $2p_i(1-p_i)$	0.50	0.47	0.33	0.33
S.D. {0,1,2} coding, $\sqrt{(2p_i(1-p_i)}$	0.70	0.68	0.57	0.57

# loci, *m*	*c = sqrt [Σ 2p_i (1-p_i)]*
4	1.25

The markers #1 to #4 have reference alleles G, A, T and G at loci 1 to 4 respectively. The individuals #1 to #4 were treated as a true base and alleles were sampled for each gamete and each locus within gamete assuming linkage equilibrium. The frequencies of the reference alleles in the gene pool forming the base were 0.3, 0.4, 0.2 and 0.3 respectively. The frequencies of the alternatives in the pedigree of 12 individuals were 13, 15, 19 and 19/24, giving means and variances as shown in the table.

X Covariate Matrix

Individual	Locus 1	Locus 2	Locus 3	Locus 4
#1	1	1	2	2
#2	1	2	1	1
#3	1	2	1	1
#4	2	1	2	2
#5	0	2	2	1
#6	0	2	2	1
#7	1	1	2	1
#8	2	1	1	2
#9	1	1	1	2
#10	1	1	1	2
#11	1	1	2	2
#12	2	0	2	2

- Coding 0,1,2

In the development in Lecture 7 it was assumed the genotypes were initially coded as covariates with values 0, 1 and 2, where 0 is the reference allele homozygote. This is **X**.

W_1 Covariate Matrix

Individual	Locus 1	Locus 2	Locus 3	Locus 4
#1	-0.08	-0.25	0.42	0.42
#2	-0.08	0.75	-0.58	-0.58
#3	-0.08	0.75	-0.58	-0.58
#4	0.92	-0.25	0.42	0.42
#5	-1.08	0.75	0.42	-0.58
#6	-1.08	0.75	0.42	-0.58
#7	-0.08	-0.25	0.42	-0.58
#8	0.92	-0.25	-0.58	0.42
#9	-0.08	-0.25	-0.58	0.42
#10	-0.08	-0.25	-0.58	0.42
#11	-0.08	-0.25	0.42	0.42
#12	0.92	-1.25	0.42	0.42

- Centred by subtracting $2p_j$
- So each column has sum =0

The next step was to centre the covariates by subtracting $2p_j$ from locus *j*. i.e. individual #3, locus #2 becomes 2-1.25 = 0.75. This is $\mathbf{W}_1$.

Since they are centred every column sums to zero.

Genomic Relationship Matrix (1)

	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12
#1	0.27	-0.42	-0.42	0.22	-0.11	-0.11	0.00	-0.05	0.00	0.00	0.27	0.38
#2	-0.42	0.79	0.79	-0.48	0.47	0.47	-0.06	-0.11	-0.06	-0.06	-0.42	-0.95
#3	-0.42	0.79	0.79	-0.48	0.47	0.47	-0.06	-0.11	-0.06	-0.06	-0.42	-0.95
#4	0.22	-0.48	-0.48	0.80	-0.79	-0.79	-0.05	0.54	-0.05	-0.05	0.22	0.96
#5	-0.11	0.47	0.47	-0.79	1.43	1.43	0.26	-1.06	-0.37	-0.37	-0.11	-1.27
#6	-0.11	0.47	0.47	-0.79	1.43	1.43	0.26	-1.06	-0.37	-0.37	-0.11	-1.27
#7	0.00	-0.06	-0.06	-0.05	0.26	0.26	0.37	-0.32	-0.27	-0.27	0.00	0.11
#8	-0.05	-0.11	-0.11	0.54	-1.06	-1.06	-0.32	0.90	0.32	0.32	-0.05	0.69
#9	0.00	-0.06	-0.06	-0.05	-0.37	-0.37	-0.27	0.32	0.37	0.37	0.00	0.11
#10	0.00	-0.06	-0.06	-0.05	-0.37	-0.37	-0.27	0.32	0.37	0.37	0.00	0.11
#11	0.27	-0.42	-0.42	0.22	-0.11	-0.11	0.00	-0.05	0.00	0.00	0.27	0.38
#12	0.38	-0.95	-0.95	0.96	-1.27	-1.27	0.11	0.69	0.11	0.11	0.38	1.76

$\mathbf{G}_1 = \mathbf{W}_1\mathbf{W}_1^T/c^2$ with $c^2 = \Sigma 2p_j(1-p_j)$

So this is Van Raden 1, $\mathbf{G}_1$.

$\mathbf{W}_2$ Covariate Matrix

Individual	Locus 1	Locus 2	Locus 3	Locus 4
#1	-0.11	-0.37	0.74	0.74
#2	-0.11	1.10	-1.02	-1.02
#3	-0.11	1.10	-1.02	-1.02
#4	1.31	-0.37	0.74	0.74
#5	-1.54	1.10	0.74	-1.02
#6	-1.54	1.10	0.74	-1.02
#7	-0.11	-0.37	0.74	-1.02
#8	1.31	-0.37	-1.02	0.74
#9	-0.11	-0.37	-1.02	0.74
#10	-0.11	-0.37	-1.02	0.74
#11	-0.11	-0.37	0.74	0.74
#12	1.31	-1.84	0.74	0.74

- Normalised
  - » divid'd by $(SD)_j$

From $\mathbf{W}_1$ to $\mathbf{W}_2$ by normalisation: the covariates were then normalised by dividing the column for marker j by its s.d. which is assumed to be $\sqrt{2p_j(1-p_j)}$. So individual #3, locus #2 becomes 0.75/0.68 ~ 1.10.

Genomic Relationship Matrix (2)

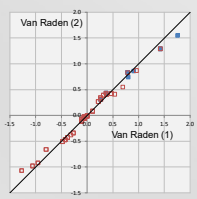
	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12
#1	0.31	-0.47	-0.47	0.27	-0.11	-0.11	-0.01	-0.06	-0.01	-0.01	0.31	0.40
#2	-0.47	0.83	0.83	-0.51	0.42	0.42	-0.03	-0.07	-0.03	-0.03	-0.47	-0.92
#3	-0.47	0.83	0.83	-0.51	0.42	0.42	-0.03	-0.07	-0.03	-0.03	-0.47	-0.92
#4	0.27	-0.51	-0.51	0.74	-0.66	-0.66	-0.06	0.41	-0.06	-0.06	0.27	0.87
#5	-0.11	0.42	0.42	-0.66	1.29	1.29	0.34	-0.98	-0.43	-0.43	-0.11	-1.07
#6	-0.11	0.42	0.42	-0.66	1.29	1.29	0.34	-0.98	-0.43	-0.43	-0.11	-1.07
#7	-0.01	-0.03	-0.03	-0.06	0.34	0.34	0.43	-0.38	-0.34	-0.34	-0.01	0.08
#8	-0.06	-0.07	-0.07	0.41	-0.98	-0.98	-0.38	0.86	0.39	0.39	-0.06	0.55
#9	-0.01	-0.03	-0.03	-0.06	-0.43	-0.43	-0.34	0.39	0.43	0.43	-0.01	0.08
#10	-0.01	-0.03	-0.03	-0.06	-0.43	-0.43	-0.34	0.39	0.43	0.43	-0.01	0.08
#11	0.31	-0.47	-0.47	0.27	-0.11	-0.11	-0.01	-0.06	-0.01	-0.01	0.31	0.40
#12	0.40	-0.92	-0.92	0.87	-1.07	-1.07	0.08	0.55	0.08	0.08	0.40	1.55

$\mathbf{G}_2 = \mathbf{W}_2\mathbf{W}_2^T/m$ where m = # loci

So this is Van Raden 2, $\mathbf{G}_2$.

Comparison (1) & (2)

- Leading diagonal
- Off-diagonal
- Not very different



The values for $\mathbf{G}_1$ and $\mathbf{G}_2$ are different, but similar.

Properties of G

- Symmetric
- **G ≠ I** in the base generation
  - » uses IBS not IBD
- Leading diagonal ≥ 0, but can be >2 and <1
  - » g_{ii} is measure of distance from 'mean' genotype.
- Off-diagonals may be < 0

However there are many differences from Wright's **A**-Matrix. Examine some properties (or lack of properties).

(i) It is symmetric $\gamma_{ij} = \gamma_{ji}$, like **A.**

(ii) In the base population (individuals #1 to #4) the γ_{ij} are not 1 if $i = j$ and 0 if $i \neq j$, this was true for **A** because of definition of IBD and the special role the base played as a reference point for IBD calculations. However **G** is based on IBS.

(iii) The γ_{ii} on the leading diagonal can be less than 1, and can also be >2 although not in this example. However they are always > 0. The value on the diagonal for individual i is proportional to the sum of squares of the values in row i of **W** (hence they must be > or = zero). Therefore low values of γ_{ii} tend to reflect the individual having 'typical' genotypes at the marker loci, closest to the mean counts. These are often the more heterozygous individuals but this is not a strict relationship: for example with low MAF the major allele homozygote is more typical. So the idea of linking the leading diagonal to heterozygosity is questionable.

(iv) The off-diagonal γ_{ij} can be < 0. This is because with the bi-allelic loci evidence of one allele being poorer is direct evidence for the other allele being better. So a tendency for two individuals to have different alleles across the set of marker loci implies *a priori* that if one individual has a better than average breeding value, the other might be predicted to be worse than average.

Genomic Relationship Matrix (1)

	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12
#1	0.27	-0.42	-0.42	0.22	-0.11	-0.11	0.00	-0.05	0.00	0.00	0.27	0.38
#2	-0.42	0.79	0.79	-0.48	0.47	0.47	-0.06	-0.11	-0.06	-0.06	-0.42	-0.95
#3	-0.42	0.79	0.79	-0.48	0.47	0.47	-0.06	-0.11	-0.06	-0.06	-0.42	-0.95
#4	0.22	-0.48	-0.48	0.80	-0.79	-0.79	-0.05	0.54	-0.05	-0.05	0.22	0.96
#5	-0.11	0.47	0.47	-0.79	1.43	1.43	0.26	-1.06	-0.37	-0.37	-0.11	-1.27
#6	-0.11	0.47	0.47	-0.79	1.43	1.43	0.26	-1.06	-0.37	-0.37	-0.11	-1.27
#7	0.00	-0.06	-0.06	-0.05	0.2							0.11
#8	-0.05	-0.11	-0.11	0.54	-1.0							0.69
#9	0.00	-0.06	-0.06	-0.05	-0.2							0.11
#10	0.00	-0.06	-0.06	-0.05	-0.37	-0.37	-0.27	0.32	0.37	0.37	0.00	0.11
#11	0.27	-0.42	-0.42	0.22	-0.11	-0.11	0.00	-0.05	0.00	0.00	0.27	0.38
#12	0.38	-0.95	-0.95	0.96	-1.27	-1.27	0.11	0.69	0.11	0.11	0.38	1.76

Sums
All = 0
Leading Diagonal = 9.55

$\mathbf{G}_1 = \mathbf{W}_1\mathbf{W}_1^T/c^2$ with $c^2 = \Sigma 2p_j(1-p_j)$

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In the handout about genetic variances and the properties of the Van Raden **G**-matrices there is proof of further properties. For example, all the elements sum to 0, because we have centred the covariates, and this constrains both the TBV (True Breeding Value) and the EBV to sum to zero for all the individuals used to construct **W** and **G**, irrespective of whether or not they have phenotypes appearing in the model.

Genomic Relationship Matrix (2)

	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12
#1	0.31	-0.47	-0.47	0.27	-0.11	-0.11	-0.01	-0.06	-0.01	-0.01	0.31	0.40
#2	-0.47	0.83	0.83	-0.51	0.42	0.42	-0.03	-0.07	-0.03	-0.03	-0.47	-0.92
#3	-0.47	0.83	0.83	-0.51	0.42	0.42	-0.03	-0.07	-0.03	-0.03	-0.47	-0.92
#4	0.27	-0.51	-0.51	0.74	-0.66	-0.66	-0.06	0.41	-0.06	-0.06	0.27	0.87
#5	-0.11	0.42	0.42	-0.66	1.29	1.29	0.34	-0.98	-0.43	-0.43	-0.11	-1.07
#6	-0.11	0.42	0.42	-0.66	1.29	1.29	0.34	-0.98	-0.43	-0.43	-0.11	-1.07
#7	-0.01	-0.03	-0.03	-0.06	0.3							0.08
#8	-0.06	-0.07	-0.07	0.41	-0.5							0.55
#9	-0.01	-0.03	-0.03	-0.06	-0.4							0.08
#10	-0.01	-0.03	-0.03	-0.06	-0.43	-0.43	-0.34	0.39	0.43	0.43	-0.01	0.08
#11	0.31	-0.47	-0.47	0.27	-0.11	-0.11	-0.01	-0.06	-0.01	-0.01	0.31	0.40
#12	0.40	-0.92	-0.92	0.87	-1.07	-1.07	0.08	0.55	0.08	0.08	0.40	1.55

Sums
All = 0
Leading Diagonal = 9.3

$\mathbf{G}_2 = \mathbf{W}_2\mathbf{W}_2^T/m$ where m = # loci

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This can be seen in both $\mathbf{G}_1$ and $\mathbf{G}_2$ although there is a little bit of rounding error for $\mathbf{G}_1$ to 2 decimal places.

Another property is that if the individuals used to construct the **W** and **G** are in HWE for all the loci used, then the leading diagonal sums to n, i.e. the average value on the leading diagonal is 1. This is not the case here as the loci are not in HWE for our 12 individuals.

Properties of G

- Symmetric
- **G** ≠ **I** in the base generation
- Leading diagonal ≥ 0, but can be >2 and <1
- Off-diagonals may be < 0
- **G** positive semi-definite
  - » elements sum to 0
  - » breeding values constrained to sum to 0
  - » strictly no inverse
  - » diagonal has mean 1 if all loci in HWE

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The implication of this is the following:

(v) **G** is positive semi-definite i.e. it is a variance (co)variance matrix but there is a linear dependency among the rows and column as the elements all sum to zero, as the sum of the breeding values has been constrained to sum to zero. There may be others, for example if there are identical twins. This means that, strictly, $\mathbf{G}^{-1}$ does not exist (see the Tutorial 7) and solutions cannot be obtained. However there are ways around this … the course does not consider these, and assumes that $\mathbf{G}^{-1}$ exists if it is needed!

Looking at Correlations

	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12
#1	1.00	-0.93	-0.93	0.56	-0.17	-0.17	-0.04	-0.11	-0.04	-0.04	1.00	0.58
#2	-0.93	1.00	1.00	-0.66	0.41	0.41	-0.04	-0.08	-0.04	-0.04	-0.93	-0.81
#3	-0.93	1.00	1.00	-0.66	0.41	0.41	-0.04	-0.08	-0.04	-0.04	-0.93	-0.81
#4	0.56	-0.66	-0.66	1.00	-0.68	-0.68	-0.10	0.52	-0.10	-0.10	0.56	0.81
#5	-0.17	0.41	0.41	-0.68	1.00	1.00	0.45	-0.93	-0.58	-0.58	-0.17	-0.75
#6	-0.17	0.41	0.41	-0.68	1.00	1.00	0.45	-0.93	-0.58	-0.58	-0.17	-0.75
#7	-0.04	-0.04	-0.04	-0.10	0.45	0.45	1.00	-0.62	-0.79	-0.79	-0.04	0.10
#8	-0.11	-0.08	-0.08	0.52	-0.93	-0.93	-0.62	1.00	0.64	0.64	-0.11	0.48
#9	-0.04	-0.04	-0.04	-0.10	-0.58	-0.58	-0.79	0.64	1.00	1.00	-0.04	0.10
#10	-0.04	-0.04	-0.04	-0.10	-0.58	-0.58	-0.79	0.64	1.00	1.00	-0.04	0.10
#11	1.00	-0.93	-0.93	0.56	-0.17	-0.17	-0.04	-0.11	-0.04	-0.04	1.00	0.58
#12	0.58	-0.81	-0.81	0.81	-0.75	-0.75	0.10	0.48	0.10	0.10	0.58	1.00

Correlation = 1?

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Looking at the correlations for identical individuals (I am not giving them a meaning). It is a quick survey showing IBD is not IBS and the consequences are different. There are 4 sets of identicals.

Examples (1)

	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12
#1	1.00	-0.93	-0.93	0.56	-0.17	-0.17	-0.04	-0.11	-0.04	-0.04	1.00	0.58
#2	-0.93	1.00	1.00	-0.66	0.41	0.41	-0.04	-0.08	-0.04	-0.04	-0.93	-0.81
#3	-0.93	1.00	1.00	-0.66	0.41	0.41	-0.04	-0.08	-0.04	-0.04	-0.93	-0.81
#4	0.56	-0.66	-0.66	1.00	-0.68	-0.68	-0.10	0.52	-0.10	-0.10	0.56	0.81
#5	-0.17	0.41	0.41	-0.68	1.00	1.00	0.45	-0.93	-0.58	-0.58	-0.17	-0.75
#6	-0.17	0.41	0.41	-0.68	1.00	1.00	0.45	-0.93	-0.58	-0.58	-0.17	-0.75
#7	-0.04	-0.04	-0.04	-0.10	0.45	0.45	1.00	-0.62	-0.79	-0.79	-0.04	0.10
#8	-0.11	-0.08	-0.08	0.52	-0.93	-0.93	-0.62	1.00	0.64	0.64	-0.11	0.48
#9	-0.04	-0.04	-0.04	-0.10	-0.58	-0.58	-0.79	0.64	1.00	1.00	-0.04	0.10
#10	-0.04	-0.04	-0.04	-0.10	-0.58	-0.58	-0.79	0.64	1.00	1.00	-0.04	0.10
#11	1.00	-0.93	-0.93	0.56	-0.17	-0.17	-0.04	-0.11	-0.04	-0.04	1.00	0.58
#12	0.58	-0.81	-0.81	0.81	-0.75	-0.75	0.10	0.48	0.10	0.10	0.58	1.00

#5 & #6 are identical twins

IBS, and IBD from base!

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#5 and #6 are identical twins and so are fully IBS, but they are also IBD with respect to the base.

Examples (2)

	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12
#1	1.00	-0.93	-0.93	0.56	-0.17	-0.17	-0.04	-0.11	-0.04	-0.04	1.00	0.58
#2	-0.93	1.00	1.00	-0.66	0.41	0.41	-0.04	-0.08	-0.04	-0.04	-0.93	-0.81
#3	-0.93	1.00	1.00	-0.66	0.41	0.41	-0.04	-0.08	-0.04	-0.04	-0.93	-0.81
#4	0.56	-0.66	-0.66	1.00	-0.68	-0.68	-0.10	0.52	-0.10	-0.10	0.56	0.81
#5	-0.17	0.41	0.41	-0.68	1.00	1.00	0.45	-0.93	-0.58	-0.58	-0.17	-0.75
#6	-0.17	0.41	0.41	-0.68	1.00	1.00	0.45	-0.93	-0.58	-0.58	-0.17	-0.75
#7	-0.04	-0.04	-0.04	-0.10	0.45	0.45	1.00	-0.62	-0.79	-0.79	-0.04	0.10
#8	-0.11	-0.08	-0.08	0.52	-0.93	-0.93	-0.62	1.00	0.64	0.64	-0.11	0.48
#9	-0.04	-0.04	-0.04	-0.10	-0.58	-0.58	-0.79	0.64	1.00	1.00	-0.04	0.10
#10	-0.04	-0.04	-0.04	-0.10	-0.58	-0.58	-0.79	0.64	1.00	1.00	-0.04	0.10
#11	1.00	-0.93	-0.93	0.56	-0.17	-0.17	-0.04	-0.11	-0.04	-0.04	1.00	0.58
#12	0.58	-0.81	-0.81	0.81	-0.75	-0.75	0.10	0.48	0.10	0.10	0.58	1.00

#2 & #3 in base!

IBS, but definitely not IBD from base!

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#2 and #3 are also identical if judged by IBS but they are both in the base generation! In fact although they are identical when judged by their diplotypes (so their genomic correlation is 1, their haplotypes are different.

With more loci this becomes very unlikely.

Examples (3)

	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12
#1	1.00	-0.93	-0.93	0.56	-0.17	-0.17	-0.04	-0.11	-0.04	-0.04	1.00	0.58
#2	-0.93	1.00	1.00	-0.66	0.41	0.41	-0.04	-0.08	-0.04	-0.04	-0.93	-0.81
#3	-0.93	1.00	1.00	-0.66	0.41	0.41	-0.04	-0.08	-0.04	-0.04	-0.93	-0.81
#4	0.56	-0.66	-0.66	1.00	-0.68	-0.68	-0.10	0.52	-0.10	-0.10	0.56	0.81
#5	-0.17	0.41	0.41	-0.68	1.00	1.00	0.45	-0.93	-0.58	-0.58	-0.17	-0.75
#6	-0.17	0.41	0.41	-0.68	1.00	1.00	0.45	-0.93	-0.58	-0.58	-0.17	-0.75
#7	-0.04	-0.04	-0.04	-0.10	0.45	0.45	1.00	-0.62	-0.79	-0.79	-0.04	0.10
#8	-0.11	-0.08	-0.08	0.52	-0.93	-0.93	-0.62	1.00	0.64	0.64	-0.11	0.48
#9	-0.04	-0.04	-0.04	-0.10	-0.58	-0.58	-0.79	0.64	1.00	1.00	-0.04	0.10
#10	-0.04	-0.04	-0.04	-0.10	-0.58	-0.58	-0.79	0.64	1.00	1.00	-0.04	0.10
#11	1.00	-0.93	-0.93	0.56	-0.17	-0.17	-0.04	-0.11	-0.04	-0.04	1.00	0.58
#12	0.58	-0.81	-0.81	0.81	-0.75	-0.75	0.10	0.48	0.10	0.10	0.58	1.00

#9 & #10 are full sibs

IBS, but not IBD at all loci

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#9 and #10 are full-sibs, are fully IBS but they are not IBD.

With more loci this becomes very unlikely, unless they are identical twins.

Examples (4)

	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12
#1	1.00	-0.93	-0.93	0.56	-0.17	-0.17	-0.04	-0.11	-0.04	-0.04	1.00	0.58
#2	-0.93	1.00	1.00	-0.66	0.41	0.41	-0.04	-0.08	-0.04	-0.04	-0.93	-0.81
#3	-0.93	1.00	1.00	-0.66	0.41	0.41	-0.04	-0.08	-0.04	-0.04	-0.93	-0.81
#4	0.56	-0.66	-0.66	1.00	-0.68	-0.68	-0.10	0.52	-0.10	-0.10	0.56	0.81
#5	-0.17	0.41	0.41	-0.68	1.00	1.00	0.45	-0.93	-0.58	-0.58	-0.17	-0.75
#6	-0.17	0.41	0.41	-0.68	1.00	1.00	0.45	-0.93	-0.58	-0.58	-0.17	-0.75
#7	-0.04	-0.04	-0.04	-0.10	0.45	0.45	1.00	-0.62	-0.79	-0.79	-0.04	0.10
#8	-0.11	-0.08	-0.08	0.52	-0.93	-0.93	-0.62	1.00	0.64	0.64	-0.11	0.48
#9	-0.04	-0.04	-0.04	-0.10	-0.58	-0.58	-0.79	0.64	1.00	1.00	-0.04	0.10
#10	-0.04	-0.04	-0.04	-0.10	-0.58	-0.58	-0.79	0.64	1.00	1.00	-0.04	0.10
#11	1.00	-0.93	-0.93	0.56	-0.17	-0.17	-0.04	-0.11	-0.04	-0.04	1.00	0.58
#12	0.58	-0.81	-0.81	0.81	-0.75	-0.75	0.10	0.48	0.10	0.10	0.58	1.00

#1 & #11

#11 has a gamete IBD to #1, but other gamete has no IBD

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#1 and #11 are IBS and IBD for one of #11's gametes, but not for the other.

Numerator Relationship Matrix

	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12
#1	1.00											
#2	.	1.00										
#3	.	.	1.00									
#4	.	.	.	1.00								
#5	0.50	0.50	.	.	1.00							
#6	0.50	0.50	.	.	0.50	1.00						
#7	0.50	0.50	.	.	0.50	0.50	1.00					
#8	0.50	0.50	.	.	0.50	0.50	0.50	1.00				
#9	.	.	0.50	0.50	.	.	.	.	1.00			
#10	.	.	0.50	0.50	.	.	.	.	0.50	1.00		
#11	0.25	0.25	0.25	0.25	0.25	0.50	0.25	0.25	0.50	0.25	1.00	
#12	0.25	0.25	0.25	0.25	0.25	0.25	0.5	0.25	0.25	0.50	0.25	1.00

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It is useful to compare **G** with **A**, To make comparisons easier only the lower triangle will be shown. Here is **A**. The pattern of values show pattern of common ancestors up to the base, and the base has matrix **I**.

All elements are either 0 or >0, so don't sum to zero; the leading diagonal is always ≥1 and ≤2; unless there are identical twins the matrix is positive definite.

Look at individuals #4 and #5. They are unrelated by our pedigree and $a_{45}=0$, but g_{45} is strongly negative (by VR1 or VR2). The reason why $g_{45}<0$ is made clear above: if they tend to carry opposing alleles identified by IBS, and the model says if one is good, then *a priori* the other may be predicted to be bad. In the pedigree model the similarity is judged by IBD, and all the base alleles are assumed to come from a different random sampling of the gene pool. If two individuals are unrelated then all their alleles are derived from a different sampling of the base generation gene pool, and so information on one does not give information on the other. If a>0, then there is IBD, so some alleles from the same random sampling will be carried by both individuals, and so information on the breeding value of one will give some information on the other.

Numerator Relationship Matrix

	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12
#1	1.00											
#2	.	1.00										
#3	.	.	1.00									
#4	.	.	.	1.00								
#5	0.50	0.50	.	.	1.00							
#6	0.50	0.50	.	.	0.50	1.00						
#7	0.50	0.50	.	.	0.50	0.50	1.00					
#8	0.50	0.50	.	.	0.50	0.50	0.50	1.00				
#9	.	.	0.50	0.50	.	.	.	.	1.00			
#10	.	.	0.50	0.50	.	.	.	.	0.50	1.00		
#11	0.25	0.25	0.25	0.25	0.25	0.50	0.25	0.25	0.50	0.25	1.00	
#12	0.25	0.25	0.25	0.25	0.25	0.25	0.5	0.25	0.25	0.50	0.25	1.00

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Look at the full sibs #5, #6, #7 and #8. **A** is an expectation conditional on the pedigree alone, so all full-sibs are the same.

Genomic Relationship Matrix

G_2

	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12
#1	0.31											
#2	-0.47	0.83										
#3	-0.47	0.83	0.83									
#4	0.27	-0.51	-0.51	0.74								
#5	-0.11	0.42	0.42	-0.66	1.29							
#6	-0.11	0.42	0.42	-0.66	1.29	1.29						
#7	-0.01	-0.03	-0.03	-0.06	0.34	0.34	0.43					
#8	-0.06	-0.07	-0.07	0.41	-0.98	-0.98	-0.38	0.86				
#9	-0.01	-0.03	-0.03	-0.06	-0.43	-0.43	-0.34	0.39	0.43			
#10	-0.01	-0.03	-0.03	-0.06	-0.43	-0.43	-0.34	0.39	0.43	0.43		
#11	0.31	-0.47	-0.47	0.27	-0.11	-0.11	-0.01	-0.06	-0.01	-0.01	0.31	
#12	0.40	-0.92	-0.92	0.87	-1.07	-1.07	0.08	0.55	0.08	0.08	0.40	1.55

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G is not predicting a specific position but the accumulation of all the QTL across the genome. #5 and #6 are identical by IBS and predictions of EBVs will be identical. #8 is judged to be strongly negatively correlated with #5 and #6 as its genotypes appear very different; look back at $\mathbf{W}_2$ all the values for #8 have the opposite sign to #5 and #6. So if it does not share the same alleles by state it must have the other alleles (remember this assumes bi-allelic), so if one is 'good' the other is 'bad' and *vice versa.* Note, with much denser SNP the half-sib and full-sib family structures become more readily discernible in **G**.

Why Use G?

- Estimation of merit with no/poor pedigree
  - » uses 'full' genetic variance, not just few loci
  - » overcomes shallow pedigree, distant relationships, or low numbers of sibs
  - » very relevant to medical or wild populations
- Does not require a map
- In pedigreed populations, discriminates among sibs cf. use of pedigree
  - » captures Mendelian sampling information

Having seen the different properties of **G** and **A** it is worth looking back at why we may want to use **G**. First of all it makes possible what was previously impossible. Where there is a lack of pedigree, or deep pedigree, there is a lack of genetic information from which to estimate genetic variance or estimate EBV. So **G,** which does not require pedigree, allows something new, relying on IBS and LD. In pedigreed populations it discriminates among sibs, and other relatives, allowing us to say these sibs are more or less alike than expected so we can capture information on Mendelian sampling.

IBD & Genomic Relationships

- If have pedigree
  - » pedigree identifies the broad relationships
  - » use the linkage analysis to identify Mendelian terms across the genome
  - » calculate $\mathbf{G}_{FG} = \int_{genome} \mathbf{A}_Q(s)\, ds$
    - » i.e. average the $\mathbf{A}_Q$ over all genome positions
    - » 'FG' after Fernando & Grossman (pioneers of MAS)

However if we have *both a pedigree and a map* we could do something different. For example, we could calculate $\mathbf{A}_Q$ for many positions on the genome and average them to get an assessment of IBD across all the genome relative to the base population for the pedigree; call this matrix $\mathbf{G}_{FG}$ (after Fernando and Grossman), *derived from linkage analysis*. Such a matrix would use all the genomic information, tracking the actual Mendelian sampling and represent a refined version of **A**, a marker informed estimate of IBD across the genome instead of the expectation of IBD - including selection effects.

$\mathbf{G}_{FG}$

	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12
#1	1.00											
#2	.	1.00										
#3	.	.	1.00									
#4	.	.	.	1.00								
#5	0.50	0.50	.	.	1.00							
#6	0.50	0.50	.	.	0.50	1.00						
#7	0.50	0.50	.	.	0.50	0.50	1.00					
#8	0.50	0.50	.	.	0.00	0.00	0.50	1.00				
#9	.	.	0.50	0.50	.	.	.	.	1.00			
#10	.	.	0.50	0.50	.	.	.	.	0.75	1.00		
#11	0.50	0.00	0.00	0.50	0.50	0.50	0.00	0.00	0.50	0.25	1.00	
#12	0.50	0.00	0.00	0.50	0.00	0.00	0.50	0.50	0.25	0.50	0.25	1.00

As mentioned in the earlier this will differentiate sibs and proliferate these differences down the pedigree at each position in the genome.

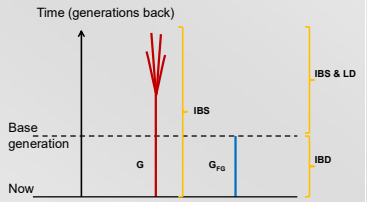
IBD & Genomic Relationships

- $\mathbf{G}_{FG}$ differs from **A** as differences among sibs proliferate down pedigree
- 'Chromosome painting' tracking segments back to founders

One difference between $\mathbf{G}_{FS}$ and **G** that **G** does not require a pedigree and a marker order on a map

The second difference is that $\mathbf{G}_{FS}$ tracks back only to the base population and then assumes all alleles are distinct. In **G** the IBS status among the alleles in the base is recognised.

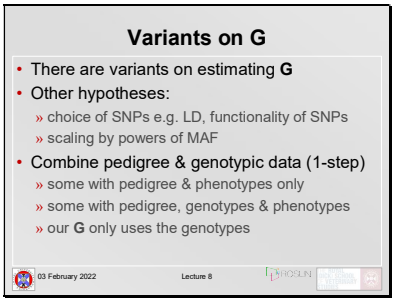
IBD, IBS & LD



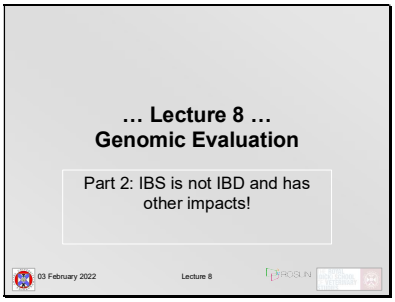
So IBS captures the IBS and the LD that was generated before the base.

G and $\mathbf{G}_{FG}$

- $\mathbf{G}_{FG}$ captures genome wide IBD
- **G** captures population wide LD and IBS
- Distinction is the set of allelic relationships among base generation animals
- Luan et al. show that in relationship-rich populations identifying IBD is the major source of info!

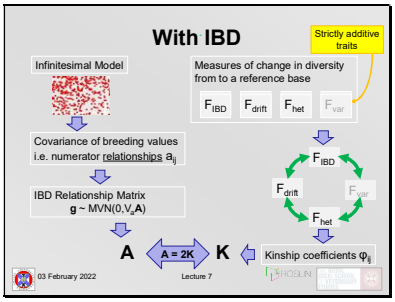


There are many variants of **G**: the choice of SNPs e.g. based on LD, or known functionality of the SNP; scaling by powers of MAF making a continuum between Van Raden 1 and 2. An important variation in breeding is where there are much large amounts of pedigree than there are genotypes, and combining the large **A**-matrix with the **G** sub-matrix, in a procedure called single-step to give what is called an **H**-matrix.

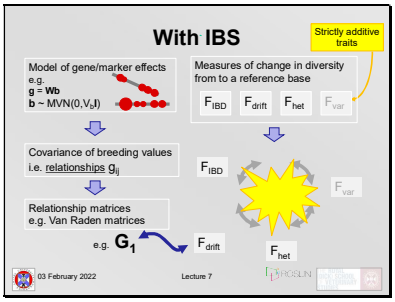


There are some more subtle consequences of moving from relationships based on IBD to those based on IBS.

Over the course of my teaching in the MSc I have stressed the idea that relationships are concerned with the (co)variances of breeding values and for that reason it is important to connect the underlying model for the breeding values themselves with the form of the relationship matrix. This is how the genomic relationships have been derived in QGM.

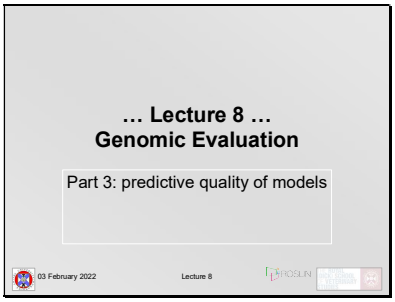


With IBD, the model leading to **A** is commonly called the infinitesimal model, and it can also be constructed as a ridge regression (see the optional 'ridge_reg.pdf') on scaled breeding vales in the base and Mendelian sampling terms of subsequent individuals. With IBD the concepts of diversity in heterozygosity, genetic drift and changes in genetic variance for additive traits coincide and can be inferred and managed through **A**.



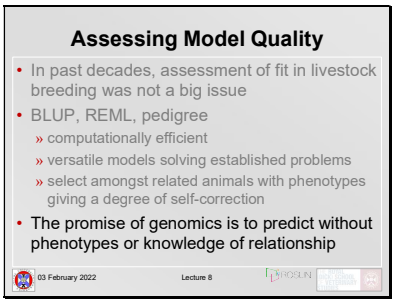
With IBS, the ridge regression models of breeding values lead to e.g. Van Raden relationship matrices **G**, which have some similarities and some differences from **A**. The perhaps unexpected outcome is that when using these matrices to infer and manage diversity this comfortable commonality of heterozygosity, drift and IBD no longer hold! Whereas $\mathbf{G}_{FG}$ which retains the use of IBD does maintain the commonality.

For further reading see Meuwissen, Sonnesson, Gebregiwergis and Woolliams (2021), Frontiers in Genetics 11:880.



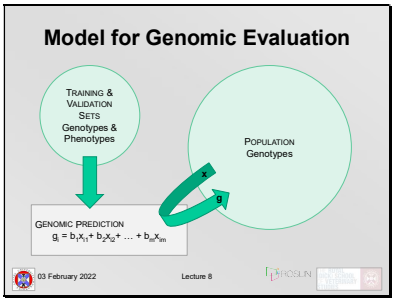
Model quality has always been important but the use of genomics has made the topic even more important. This is because genomics offers much more powerful opportunities beyond the safety of being anchored to pedigree and phenotype.

Until now our idea of accuracy has been based on assuming the model and its parameters are correct. If the model is well-established that may (with regular review!) be OK, but it may be very (dangerously) misleading if the model is somehow very wrong either in its structure or parameterisation.

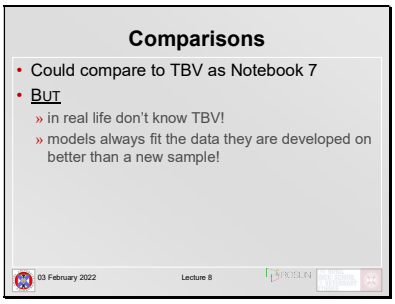


When anchored to pedigree and phenotypes there is an element of self-correction, as selection usually involved individuals with phenotypes, even if information from relatives is included. Therefore one had some quality assurance, and self-correction if errors were made: there was a direct and feedback in the prediction as in the next set of predictions those phenotypes would be included in the data.

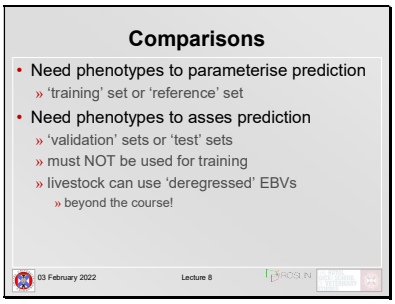
However one of the promised benefits of genomics is the prediction of breeding value or genetic risk in the absence of phenotypes. If important decisions are to be made, e.g. in medical advice, it is clear that some assessment of quality is required before implementation.



At the heart of this benefit is the following concept: (i) a training set of data is obtained where both phenotypes and genotypes and this is used to develop a predictive model of breeding value or genetic merit as a function of the genotypes; (ii) a validation set is used to test the predictions, and this too has phenotypes and genotypes, with the important proviso that this data was *not* used to develop the predictions; and (iii), if successfully validated this prediction is rolled out throughout the population based only on genotype. This in phase (iii) genetic predictions of complex traits (e.g. disease liability, expensive to measure traits) leading to critical decisions are made without a need for observation or measurement or knowing anything of pedigree. Hence the need for quality assurance!

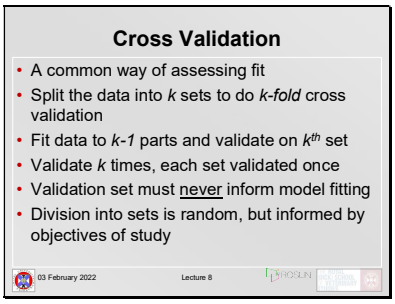


We cannot compare with TBV as in simulations! It is also the case that models derived from the training set fit the training set better than they would an independent sample – model fitting finds the parameters that best fit the training data.



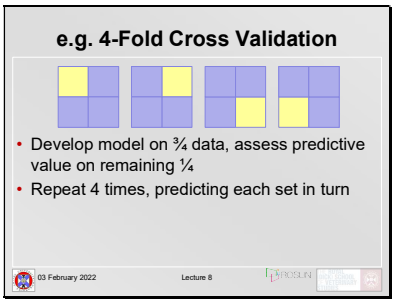
We must use phenotypes to validate the model as well as parameterise it. In breeding we can sometimes use what are called de-regressed EBVs which convert EBVs back to 'phenotypes' as an efficient alternative to using phenotypes, but this is not a common situation.

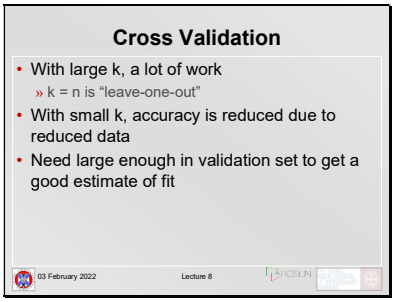
Hence we arrive at the training (or reference) set of phenotypes to estimate the parameters, and a validation (or test) set to see how good the predictions are. *The golden rule for the validation set is that it must not have played a role in the training of the model.*



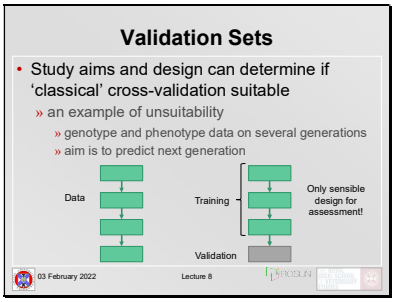
So how might one assess prediction quality from the data we have? One method is cross-validation, where the data is split into *k* subsets. Then each of the *k* subsets is used as a *validation* set using predictions derived *only* from the other (*k*-1) subsets. The set of animals used to develop the predictions is called the *training* set.

In doing this the subset that is left out must not contribute to any part of the model fitting. Division into the subsets is at random for classical cross-validation, but should always be informed by the purpose of the prediction.



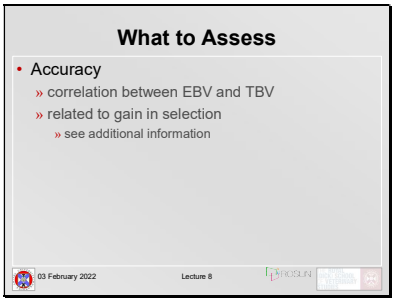


The value of *k* can be chosen. Large *k* (*k*=*n* is called 'leave one out') means fitting the model many times and can be very time consuming. The outcomes are very correlated as the training data is very similar each time. On the other hand for very small *k* (e.g. 2-fold) the model fitting is then made with a very reduced data set, and a pessimistic view of accuracy is obtained. (In all cross-validation the prediction accuracy is reduced to some degree because the training data is less than the full data). However we may have models that help us scale the accuracy for a given number of records. A further consideration is that the validation set should be large enough to show some characteristics of the fit of the model.

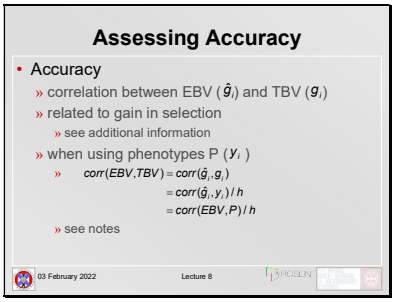


Classical k-fold validation may not always be suitable. For example if one has data over several generations and the objective is to predict the youngest cohort in a population, it does not make much sense dividing the population within each generations. In such an example the only meaningful validation is predicting the last cohort available from all the older cohorts available.

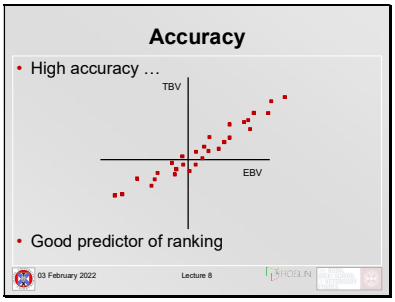
So when carrying out validation always ask the question what is the objective of the prediction, i.e. what will it be used for, as this may influence both the division into training and validation sets and what is measured in the validation set to assess quality.



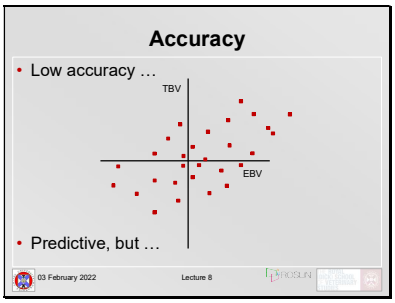
An important criteria for a prediction is its accuracy. In quantitative genetics its use is more precise: it is the correlation between the EBV and the TBV. It is particularly important in breeding as ΔG is directly related to the accuracy (see Breeders Equation pdf), but its relevance to other branches of quantitative genetics is obvious!



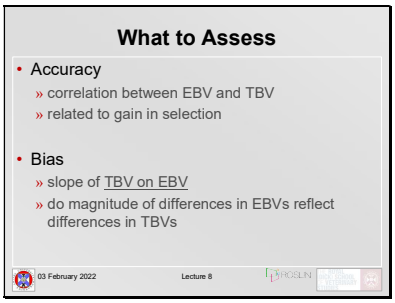
When using phenotypes one must remember that unless $h^2 = 1$, even a perfect estimate of the TBV will not have correlation 1 with the phenotype. The maximum correlation that can be obtained is $corr(y_i, g_i) = cov(y_i, g_i)/[\sigma_A\sigma_P] = \sigma_A^2/[\sigma_A\sigma_P] = \sigma_A/\sigma_P = h$, i.e. the square root of the heritability, which is the accuracy of mass selection. The correlation with phenotype is interesting, but to obtain the accuracy of the EBV the outcome is scaled by dividing by h as an estimate of the accuracy of predicting the TBV.



High accuracy, ~ 0.9, the detailed ranking is also quite reliable.

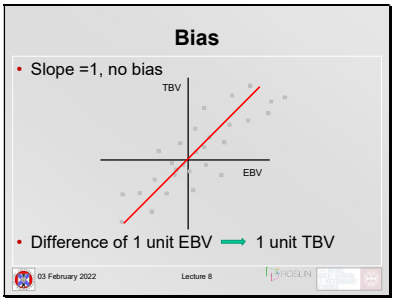


Low accuracy, say ~ 0.3. Some predictive power, but ranking unreliable, although one would expect to make slow selection progress (equivalent to mass selection with $h^2 = 0.09$).



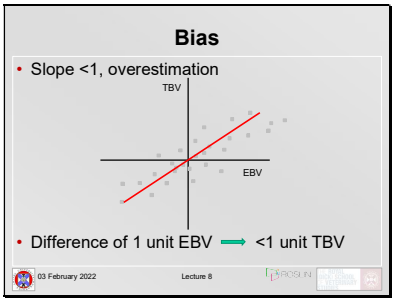
Another parameter of interest is bias, which is the slope of the TBV regressed on the EBV. This is of considerable importance if one is comparing EBV with different accuracies.

Providing (i) we assume the phenotype $y_i = g_i + e_i$ (which we do!), (ii) the individuals used for validation have played no role in forming the predictions of g_i (fundamental in cross-validation), and (iii) our model is adequate in removing any systematic environmental covariance, then $cov(e_i, \hat{g}_i) = 0$. Therefore, the slope of the phenotype y_i on the EBV $\hat{g}_i$ is the same as slope of the TBV on the EBV, as $cov(y_i, \hat{g}_i)/var(\hat{g}_i) = cov(g_i, \hat{g}_i)/var(\hat{g}_i)$; i.e. in cross-validation regress the phenotypes on the TBV and interpret the slope as for TBVs.



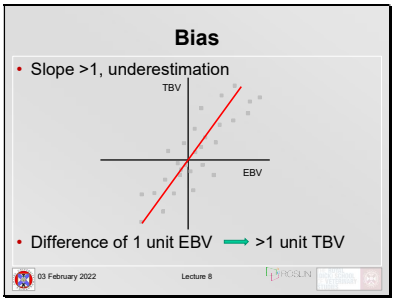
An unbiased predictor has a slope of 1, then a difference of *x* units in EBV would be expected to represent a difference of *x* units in TBV.

Note the 'scaled phenotype' EBV of mass selection is unbiased: $\hat{g}_i = h^2 y_i$, $cov(g_i, \hat{g}_i) = h^2 var(g_i)$ (as above), $var(\hat{g}_i) = var(h^2 y_i)$ $= h^4 var(y_i)$; so the slope is $cov(g_i, \widehat{g_i}) / var(\widehat{g_i})$ $= (h^2 / h^4)(var(g_i) / var(y_i) = h^2 h^2 / h^4 = 1$.

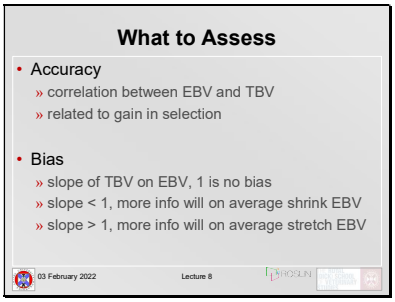


If slope is <1 the scale of the EBV would be expected to shrink as more information is accumulated as a difference of *x* units in EBV represents <*x* units in TBV; the EBVs are overestimating true differences.

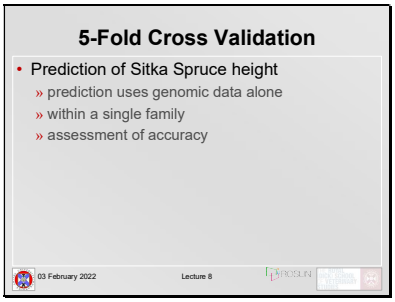
Look back to the Lecture 5 … the least squares estimates are biased: they overestimate true differences!



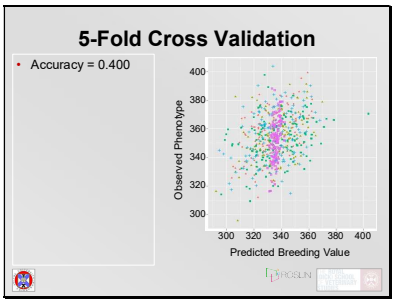
If slope is >1 the scale of the EBV would be expected to stretch as more information is accumulated as a difference of *x* units in EBV represents >*x* units in TBV; the EBVs are underestimating true differences.



It is possible to carry out a statistical test of bias using the slope of the estimated regression and its s.e. with H_0: slope =1 with alternative H_1: slope ≠1, which is two-sided.



The following is an example of classical 5-fold cross-validation EBVs within a single large full-sib family of Sitka Spruce. The division into 5 equal sets was completely at random



The correlations of predicted EBV with phenotype were calculated. The correlations for the five sets were: 0.455, 0.426, 0.317, 0.335 and 0.447. The average correlation with phenotype was 0.4. As well as the average correlation over the *k* folds, it is important to calculate the range, or s.d., as an assessment of the uncertainty.

The heritability within families is $\frac{1}{2}h^2 / (1-\frac{1}{2}h^2)$ and if this value is assumed to be 0.36 from previous estimates of h^2, an estimate of the accuracy of predicting the TBV within the family is $0.4/\sqrt{0.36} = 0.4/0.6 = 0.67$.

… Lecture 8 …
Genomic Evaluation

Part 4: using ROC curves to assess models

In medicine or veterinary medicine, one tends to have disease data. Studies of this type often assess quality by ROC, Response Operator Curves. A genomic predictor can be obtained which quantitatively predicts the genetic risk of being 'Diseased' or 'Healthy'. However such predictors are rarely perfect (and can't be perfect unless the occurrence of disease is entirely genetically determined). See the description of ROC curves uploaded on Learn.

What to Assess

- In medicine, often disease data
- Continuous predictor of disease
  - » classify as diseased above a given threshold
  - » Se, sensitivity, P(Predictor Diseased given Diseased)
    - » P(Predictor TRUE given Diseased)
  - » Sp, specificity, P(Predictor Healthy given Healthy)
    - » P(Predictor TRUE given Healthy)
  - » key epidemiological terms
    - » not the same as false positive rates
    - » not the same as P(Predictor TRUE) as this will depend on prevalence i.e. fraction of population Diseased.

So the genetic predictor is a continuous function of liability and this may need to be interpreted as either predicting disease or no disease. In many tests a threshold is then introduced below which no action is taken and above which interventions such as treatments or further tests or monitoring (all of which involve more cost, more work, more stress to the patient).

What to Assess

	Predicted Healthy	Predicted Diseased	Total
Healthy	N_{H0}	N_{H1}	N_H
Diseased	N_{D0}	N_{D1}	N_D
Total	N_0	N_1	N

$Se = N_{D1} / N_D$
$Sp = N_{H0} / N_H$

Key epidemiological terms related to a test are Specificity (Sp) and Sensitivity (Se). They are not false positives and false negatives, beware: Sp and Se look at the test conditional on the true status of the individuals, NOT conditional on the test outcome!

Se is P(Test outcome is 'Diseased' | True state is 'Diseased')

Sp is P(Test outcome is 'Healthy' | True state is 'Healthy')

Disease Risk Predictor

t*: Se = 2/3, Sp = 4/5

Predictor of Liability, T

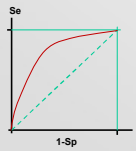
Perfect tests are often not possible

The threshold of a continuous predictor used to classify 'Diseased' and 'Healthy' can be varied. For each threshold one can calculate the sensitivity Se and the specificity Sp.

This can then be plotted as a ROC curve, a 'Receiver Operator Characteristic' curve, with 1-Sp on the x axis and Se on the y-axis.

What to Assess

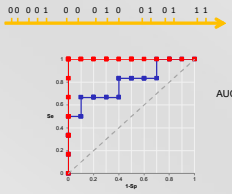
- Predictor of disease
  - » Se, sensitivity … P(Predictor True given Diseased)
  - » Sp, specificity … P(Predictor True given Healthy)
  - » Receiver Operator Characteristic (ROC) curves
    - » change threshold t*
    - » calculate Se, Sp
    - » judged by area under curve (AUC)
    - » ideally 1, random is ½



The test is then assessed by the area underneath the ROC curve (the AUC, Area Under Curve)

If the test is 'perfect' then there is a threshold that can be used to divide the distribution so that all cases and healthy individuals are predicted correctly, and AUC=1. It is not surprising that most tests are imperfect, and the AUC is< 1. A random predictor follows the straight line Se = 1-Sp, so the area under the curve is ½.

Imperfect ROC



With more points the steps become smaller and the curve becomes smoother. For more information see work by Wray et al.

Different medical interventions might be justified with different values of AUC. For example, an imperfect screening test which is minimally invasive and (say) AUC > 0.7 might be justified, with a positive outcome leading to further screening which may be more predictive but which is more invasive. Genetics cannot deliver an AUC of 1 unless the disease is completely genetically determined.

… Lecture 8 …
Genomic Evaluation

Part 5: concluding notes on predictive quality

It is easy to think because it is ‘genomic’ it is better than just pedigree, assuming it is available

Pedigree

- Pros
  - i. relatively simple model
  - ii. computationally efficient
- Cons
  - i. need to record pedigree
  - ii. need (close) relatives recorded
  - iii. accuracy at birth < 0.71

It is easy to think because it is ‘genomic’ it is better than just pedigree, assuming it is available.

Pedigree has its pros in that it is simple and computationally efficient. Its cons are that it is anchored to the pedigree and phenotypes, and is unable to capture Mendelian sampling without a phenotype on itself or offspring. So at birth the accuracy of prediction is at best 0.71, and typically much smaller.

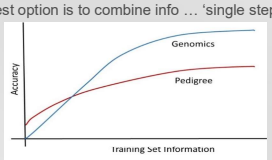
Genomics

- Pros
  - i. no need for pedigree
  - ii. no need for close relatives
  - iii. (potentially) high accuracy at birth
- Cons
  - i. needs large training set
  - ii. require large # SNP
  - iii. large # records computationally heavy

Genomics breaks free of these constraints arising from pedigree, but it needs large training sets with genotypes and phenotypes, using a large and robust set of SNPs, and can be computationally demanding.

Genomic vs Pedigree

- Without pedigree genomics is the only option!
  - » don’t just assume genomic must be better!
  - » best option is to combine info … ‘single step’



Therefore pedigree models, if available, can perform better than genomic models if the training data is small or flawed. However as the training data accumulates one expects the genomic approaches to become robust and demonstrate their superiority.

In practice it is not either/or. In genetic evaluation all information is used and so there are ways (‘single-step’, as mentioned above) to combine the different sources of information – not covered in QGM.

Accuracies

- Model based accuracy

$$r_i = \sqrt{[1-(s.e.)^2 / (a_{ii}V_A)]}$$

- Validation accuracy

$$r = corr(\hat{g}, P) / h$$

We now have two measures of accuracy. The first coming from the s.e.s calculated within our model, which assume the model and the parameters used are correct. The second comes from the comparison of predictions with phenotypes in a validation set.

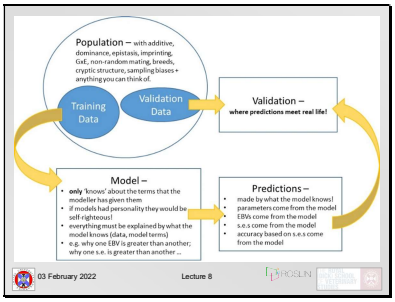
They are different!

Model vs Validation Accuracy

- Model Accuracy
  - » parameters based on all data
  - » readily available
  - » customises to individuals with differing information
  - » assumes model is correct
  - » model is wrong
- Validation Accuracy
  - » parameters estimated with subset
  - » separate design
  - » doesn’t easily extend to variable information
  - » does not assume model correct
  - » real life

Here are some pros (green) and cons (red). If the model is well-tested and well-established the ‘model’ accuracy will be reasonable, but if something new or unknown is being done then these may be misleading.

When using validation data (as presented here) then: (i) all EBVs derived from the training set, and (ii) validation data on the individuals in the validation set are treated as being independent and equal. This problem can be addressed to some degree in the validation procedures and models, but it requires careful consideration. When assuming the model is correct this problem is accounted for.



Even if the model and its implementation in a population is well-established, and the model accuracy is used routinely, my advice is to test it with a validation set from time to time!